Abstract 733

Noncanonical Proteins Detected in the Proteome and MHC-I Immunopeptidome Represent Broadly Non-Overlapping Universes

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Introduction

● Noncanonical proteins derive both from translating ostensibly non-coding ORFs[1] and frameshift of canonical proteins.

● Accumulating evidence suggests that noncanonical proteins often possess critical and diverse cellular features when compared to classic ORFs: they are shorter, have lower transcription and translation rates, commonly initiate translation on near-cognate codons (i.e., differ from AUG by a single nucleotide) and are predicted to be less stable *in-vivo*. (1-6)

● To date, Ribo-seq[2] and MS[3] analyses have been powerful tools to find such new ORFs. (4)(7-9)

● The short length and low abundance of noncanonical proteins makes their detection in whole cell extracts by shotgun MS analyses challenging. However, MHC-I[4] serves as a sink for peptides whose source protein translation would otherwise be invisible to MS due to their rapid degradation.

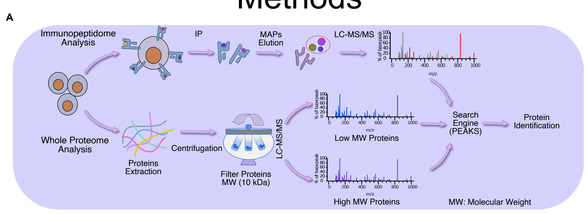
Hypothesis: Immunopeptidomics and whole proteome analyses using ribosome profiling and MS will allow broaden the landscape of **noncanonical proteins**.

Objectives

1) To develop a proteogenomic approach to broaden the identification of noncanonical proteins.

2) To study the properties of the noncanonical proteins identified on immunopeptidomics and whole proteome analysis to compare them to canonical proteins.

Methods



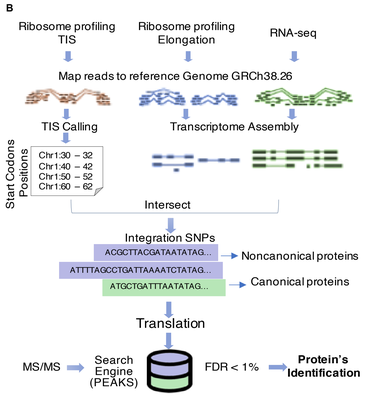


Figure 1.

(A) Workflow for detection of noncanoncial proteins:

Immunopetidome and whole proteome analysis.

(B) General overview of the workflow used to generate sample-specific databases containing active canonical and noncanonical translations based on Ribo-Seq data.

Immunopeptidome Analysis

7,045 DLBCLs-derived MAPs derive from canonical and noncanonical genomic regions

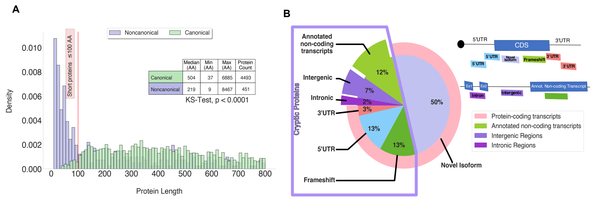


Figure 2. (A) Length distribution of canonical and cryptic noncanonical MAPs source proteins. Proteins with a length >800 amino acids are not displayed. **(B)** Noncanonical proteins derive from coding and non-coding transcripts. Diagram illustrating how various types of transcripts were designated as a function of their genomic location.

Divergent properties of noncanonical and canonical MAP source proteins

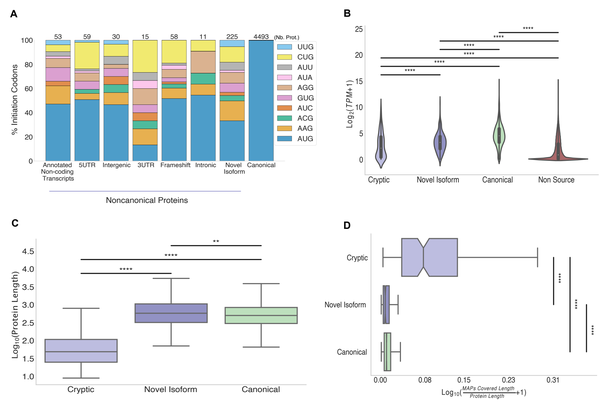


Figure 3. Properties of MAP source proteins. (A) More than a half of the noncanonical MAPs source proteins (60%) initiated at a near-cognate codon. **(B)** Transcript expression level of transcripts source of MAPs source proteins and non-source proteins. **(C)** Length distribution of MAPs source proteins for each category: cryptic, novel isoform and canonical. Median length in cryptic (49 amino acids), canonical (504 amino acids) and novel isoform (582 amino acids). **(D)** Cryptic proteins are proficient to generate MAPs.

Whole-Proteome Analysis

The vast majority of proteins identified in the low MW fraction were cryptic proteins, while canonical proteins dominated the high MW fraction

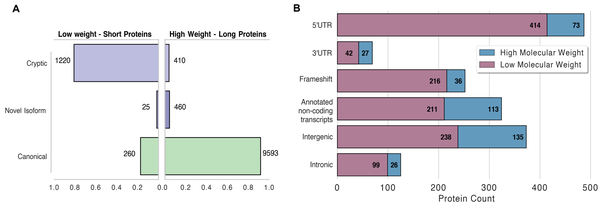


Figure 4. Canonical and cryptic proteins detected in tryptic digests of whole cell extracts. (A) Low weight fraction is enriched in cryptic proteins whereas high weight fraction is enriched in canonical proteins. **(B)** Genomic origin of cryptic proteins identified in the whole proteome extracts.

Whole cell proteome and immunopeptidome have a limited overlap

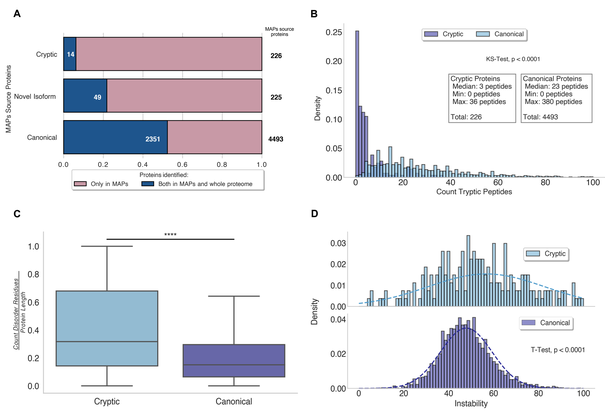


Figure 5. Cryptic proteins are disordered and unstable. (A) Bar plot depicting the total number of proteins identified in the immunopeptidome (pink bars) and the overlap with proteins detected in the whole proteome (blue bars). **(B)** Distribution of the number of predicted tryptic peptides per MAP source protein. **(C)** Cryptic proteins contain significantly more disordered residues than canonical proteins. **(D)** Cryptic proteins are less stable *in-vivo*. Proteins with instability indexes over 40 are predicted to be less stable.

Features of Noncanonical Proteins

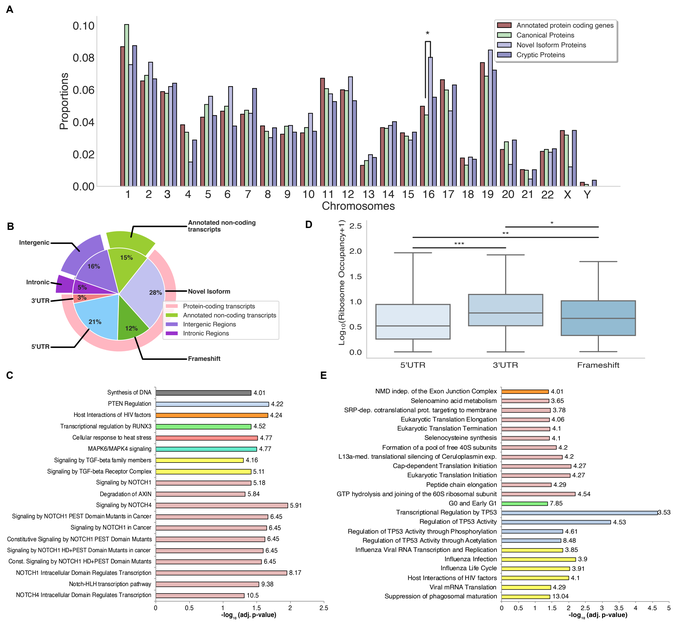


Figure 6. Chromosomal origin and function of noncanonical proteins. (A) Newly identified proteins derive from all chromosomes. Chromosome 16 generated more novel isoform proteins in DLBCL cell lines. **(B)** Genomic origins of the whole set of newly identified proteins. **(C)** Novel isoform proteins derive from genes that regulate pathways commonly perturbed in DLBCL and other cancers. **(D)** Ribosome occupancy of the canonical coding sequence (CDS) of a gene that produced a cryptic protein via frameshift, 5'UTR or 3'UTR translation. **(E)** 5'UTR cryptic proteins regulate the translation of canonical proteins involved in transcription, translation and anti-viral responses.

Summary

● We have developed a proteogenomic method to identify unannotated proteins whose peptides are detected in the whole cell proteome and immunopeptidome. Our approach, which integrates RNA-Seq, Ribo-Seq and MS data, identified 2,503 new noncanonical human proteins expressed from all chromosomes: 1,842 cryptic proteins (72%) and 661 new protein isoforms (28%).

● Cryptic transcripts generated MAPs ~5 fold more efficiently than canonical transcripts. The most plausible explanation is that cryptic proteins are rapidly degraded because they are disordered and unstable.

● The global proteome, largely consisting of stable proteins, has a limited overlap with the immunopeptidome. Remarkably, only 6% of cryptic MAP source proteins were detected in tryptic digests of whole cell extracts.

● 5'UTR cryptic proteins act as cis-regulatory mechanism that would prevent canonical translations.

● Underlying genomic aberrations may impact the noncanonical translation landscape by inducing an increase in novel isoform or cryptic proteins. How this affects the presentation of tumor specific antigens that can be targeted for immunotherapy will be explored in further studies.

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Key Words

1. Open Reading Frames
2. Ribosome Profiling
3. Mass spectrometry
4. Major Histocompatibility Complex I
5. MHC-I Associated Peptides
6. Diffuse Large B cell Lymphomas

Acknowledgements

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